

Quasi-isospectrality of manifolds and Schrödinger Operators

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IV. Schrödinger Operators on an interval - Existence

Explicit construction by Bilotta, Morassi & Turco (1993)

Method

- also found in the book by Pöschel-Taubowitz (1987)
- Based on

- * the single commutation method

- * the double commutation method

developed by Gel'fand, Leviatan, Darboux, Crum, Deift, Gesztesy-Tel'sch,

to insert and remove an eigenvalue in a spectral gap.

Let $q \in L^2[0,1]$, $\mathcal{Z}_q$: operator associated to the problem

$$\left[\begin{array}{l} -y'' + qy = \lambda y \quad \lambda \in (0,1) \\ y(0) = 0 = y(1) \end{array} \right.$$

Defⁿ. Two potentials $q, p \in L^2[0,1]$ are quasi-isospectral if the operator $\mathcal{Z}_q$ and $\mathcal{Z}_p$ are.

Existence: Given q , construct p quasi-isospectral to q .

Statement : (BMT, PT): let $q \in L^2[0,1]$, $\text{Spec}(Z_q) = \{\lambda_k(q)\}_{k \geq 1}$

set $\lambda_0 = 0$, let $t \in \mathbb{R}$, $n \geq 1$ be such that

$$\lambda_{n-1}(q) < \lambda_n(q) + t < \lambda_{n+1}(q)$$

Then $\exists p \in L^2[0,1] : \lambda_m(p) = \lambda_m(q) + t \delta_{m,n} \quad \forall m \geq 1$

In order to build p we need Darboux's lemma

Lemma (Darboux) let $\mu \in \mathbb{R}$ be fixed, and

let $f \neq 0$ s.th. $-f'' + qf = \lambda f$, $\lambda \neq \mu$

let $g \neq 0$ s.th. $-g'' + qg = \mu g$

then $h = \frac{1}{g} (gf' - g'f)$ satisfies

$$-h'' + \left(q - 2 \frac{d^2}{dx^2} \log(g)\right) h = \lambda h \quad (*)$$

Moreover, if $\lambda = \mu$ the general solution of $(*)$ is

$$\frac{1}{g} \left(a + b \int_0^x g^2(s) ds \right) \quad \text{for } a, b \in \mathbb{R}$$

Construction of p (simplified)

Consider $\{e_m\}_{m \geq 1}$ $Z_q e_m = \lambda_m e_m$ $e_m(0) = 0$
 $e_m'(0) = 1$

let n and t be as above, consider

$$y_1 = y_1(x, q, \lambda + t) : -y_1'' + q y_1 = (\lambda + t) y_1$$

$$y_1(0) = 1, y_1'(0) = 0$$

$\leadsto$ constructs w_t that satisfies

$$-w_t'' + q w_t = (\lambda + t) w_t$$

$$w_t(0) = 1 \quad \text{and} \quad w_t(1) = y_1(1, q, \lambda)$$

Now there are 3 Steps

Step 1 DL with $f = w_t$, $g = \ell_n$, $\boxed{p_{n,t} = w_t \ell'_n - w'_t \ell_n}$

then $h = \frac{p_{n,t}}{\ell_n}$ satisfies $-h'' + \hat{q} h = (\lambda + t) h$

Step 2 DL with $f = \ell_m$ and $g = \ell_n$ $\left(\hat{q} = q - 2 \frac{d^2}{dx^2} \log(\ell_n) \right)$

then ℓ : $-\ell'' + \hat{q} \ell = \lambda_m \ell$

Step 3 DL with $f = \ell$ and $g = h$

$\Rightarrow K_{m,t}$ that satisfies $-K''_{m,t} + p K_{m,t} = \lambda_m K_{m,t}$

with $\boxed{p = q - 2 \frac{d^2}{dx^2} \log(p_{n,t})}$

if $m=n$ $K_{n,t} = \frac{\ell_n}{p_{n,t}}$

To notice: $p_{n,t} > 0$ & $p_{n,t} \in L^2$

$K_{m,n}$ are well defined, non-trivial and satisfy the bdcary condition.

Example $q=0$, $-y'' = \lambda y$ $y(0)=0$ $y(1)=0$ $\text{Spec}(\mathcal{L}_0) = \text{Spec}(\Delta) = \{n^2 \pi^2\}_{n \geq 1}$

$n \geq 2$ $(n-1)^2 \pi^2 < \mu < (n+1)^2 \pi^2$, $\mu = n^2 \pi^2 + t$

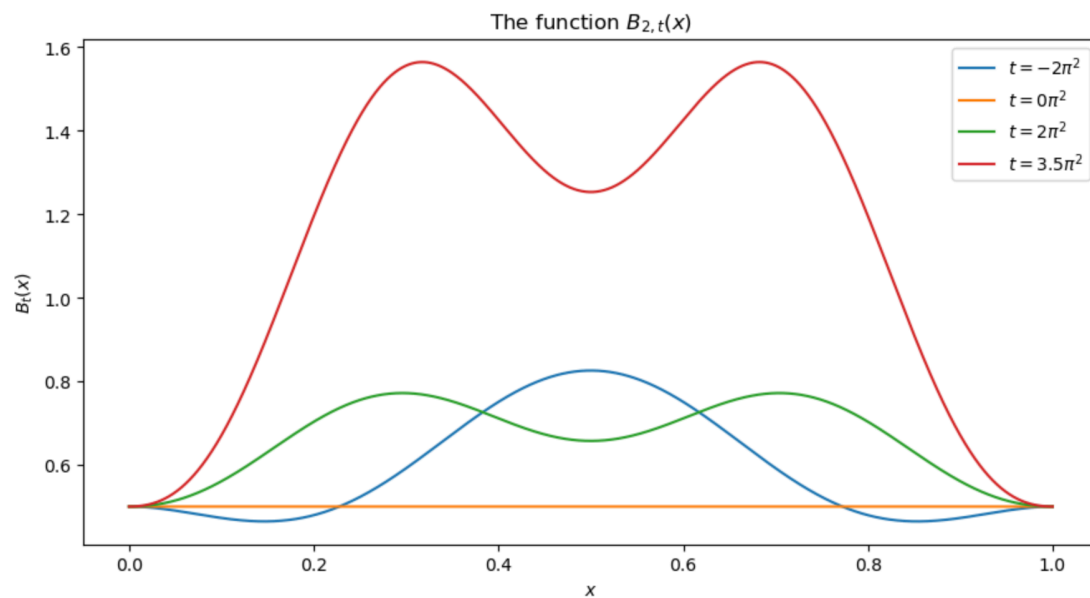
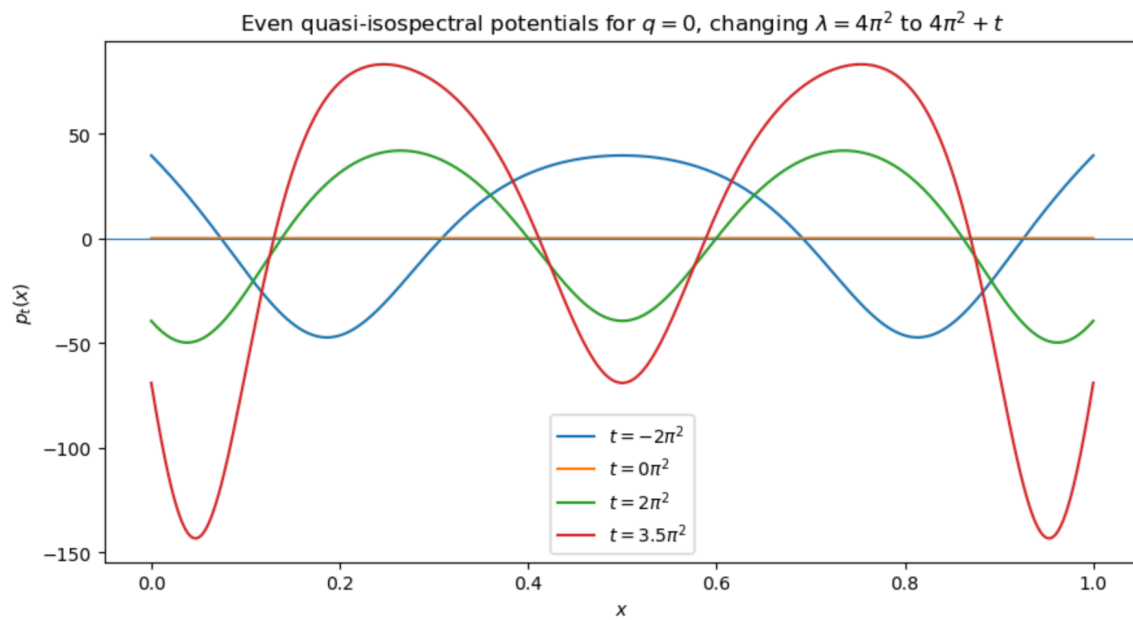
Using the method we obtain a family of even potentials

that are quasi-isospectral to $q=0$ $\hookrightarrow$ respect to $x=1/2$

$$p_{n,t} = - \frac{d^2}{dx^2} \log(p_{n,t})$$

$$p_{n,t} = \frac{1}{\sin \sqrt{\mu}} \left(\cos(n\pi x) \left(\sin(\sqrt{\mu}(1-x)) + (-1)^n \sin(\sqrt{\mu} x) \right) + \frac{\sqrt{\mu}}{n\pi} \sin(n\pi x) \left(\cos(\sqrt{\mu}(1-x)) + (-1)^n \cos(\mu x) \right) \right) \quad \left. \vphantom{p_{n,t}} \right\} \text{ even}$$

Graphics made in Python with assistance of ChatGPT



Vielen Dank für Ihre Aufmerksamkeit!